

Multiple Choice (10 marks)

1. A DNA molecule has 180 base pairs and is 20 percent adenine. How many cytosine nucleotides are present in this molecule of DNA?
 - (a) 108 nucleotides
 - (b) 120 nucleotides
 - (c) 150 nucleotides
 - (d) 144 nucleotides
 - (e) 60 nucleotides
2. What hormones are involved with the changes that occur in a pubescent female and male?
 - (a) Cortisol, aldosterone, and adrenal androgens
 - (b) Insulin, glucagon, and somatostatin
 - (c) Cholecystokinin, ghrelin, and leptin
 - (d) Vasopressin, erythropoietin, and renin
 - (e) Epinephrine, norepinephrine, and dopamine
3. About 0.5 percent of the American population have epilepsy, the falling sickness. Discusses the genetics of this disease.
 - (a) Epilepsy is the result of a chromosomal aneuploidy, similar to Down syndrome.
 - (b) Epilepsy is always inherited in an X-linked recessive pattern.
 - (c) Epilepsy is a dominant trait.
 - (d) Epilepsy is caused by an autosomal dominant trait.
 - (e) Epilepsy is due to an autosomal recessive trait.
4. In the formation of the earliest cells, which of the following components most likely arose first?
 - (a) Lysosome
 - (b) Chloroplast
 - (c) Ribosome
 - (d) Golgi apparatus
 - (e) Plasma membrane
5. Which of the following characteristics would allow you to distinguish a prokaryotic cell from an animal cell?

- (a) Centrioles
- (b) Chloroplasts
- (c) Lysosomes
- (d) Cell wall
- (e) Mitochondria

True / False (10 marks)

Answer True or False

6. True or False: Polygenic traits are characteristics that are influenced by multiple genes, resulting in a wide range of phenotypes.
7. True or False: Cycads possess the ability to photosynthesize at night, which is a common trait among many plant species.
8. True or False: Complete equilibrium in a gene pool can always be achieved under ideal conditions with no external influences.
9. True or False: In chick embryos, the yolk sac is the extraembryonic membrane that provides nourishment to the developing fetus.
10. True or False: Distantly related organisms can have different rates of mutation due to varying evolutionary pressures and environmental factors.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the number of chromatids present during meiosis in an organism with a diploid number of 8, specifically in the tetrad stage, late telophase of the first meiotic division, and metaphase of the second meiotic division.
12. Discuss how temperature, precipitation, soil chemistry, and decomposer types influence the characteristics and diversity of organisms within an ecosystem.

End of Paper